

AQUEDUCTS IN THE COPPER SCROLL*

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Introduction

Several fortresses were built during the Second Temple period in the region of Qumran, within 20 km of the site (see Fig. 6.1).¹ In the Copper Scroll the aqueducts that brought water to Khirbet Qumran as well as to some of those fortresses are mentioned.²

One may assume that the author of the Copper Scroll was familiar with these royal fortresses. During the first century CE these fortresses were guarded by Roman soldiers. Aqueducts led water to all these fortresses. In every fortress cisterns were carved in order to store the water. Those aqueducts were quite notable, and as such were used by the author of the Copper Scroll as a reference to the hidden treasures.

This article is divided into two sections. In the first section I would like to describe briefly four aqueducts that were built in the area of Qumran. In the second part I discuss the parts of the Copper Scroll that are related to aqueducts.

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1. Y. Tzafrir, 'The Desert Fortresses of Judaea in the Second Temple Period', *The Jerusalem Cathedra* 2 (1982), pp. 120-45; G. Garbrecht and J. Peleg, 'Die Wasserversorgung geschichtlicher Wüstenfestungen am Jordantal', *Antike Welt* 20.2 (1989), pp. 2-20; G. Garbrecht and Y. Peleg, 'The Water Supply of the Desert Fortresses in the Jordan Valley', *BA* 57 (1994), pp. 161-70.

2. I would like to thank D. Amit for his comments on some of the issues discussed in this paper.



Fig. 6.1. *Map of all the fortresses in the Judean desert*
(Taken from Tzafrir, 'The Desert Fortresses' [see n. 1])

The Aqueducts in the Qumran Area

The Aqueduct to Qumran

A short aqueduct of c. 0.5 km in length, starting under a high waterfall in Wadi Qumran, brought water to Khirbet Qumran (Fig. 6.2).³ This aqueduct consisted of two parts. The first part was built in Wadi Qumran, keeping its height at a uniform level by means of tunnels and supporting walls. The second part of the aqueduct was built on the plateau.

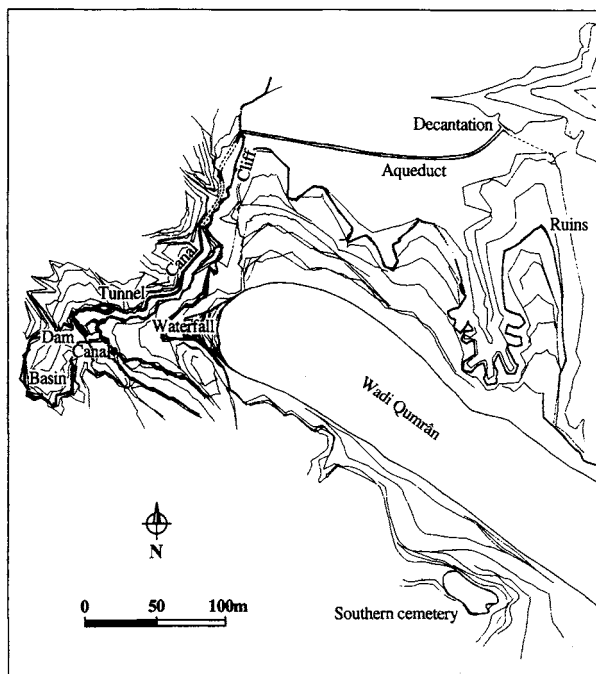


Fig. 6.2. *Map of the aqueduct of Qumran*

(Taken from de Vaux, *Fouilles de Khirbet Qumrân et de Ain Feshkha* [see n. 3])

3. E.W.G. Masterman, 'Notes on Some Ruins and a Rock-cut Aqueduct in Wadi Kumran', *PEQ* 35 (1903), pp. 265-67; R. de Vaux, 'Fouilles de Khirbet Qumran', *RB* 63 (1956), pp. 538-40 [573]; S. Schultz, 'Chirbet Kumran, En Feschcha und die Bukea', *ZDPV* 76 (1960), pp. 53-58; Z. Ilan and D. Amit, 'The Aqueduct of Qumran', in D. Amit, Y. Hirschfeld and J. Patrich (eds.), *The Aqueducts of Ancient Palestine* (Jerusalem: Yad Izhak Ben-Zvi, 1989), pp. 283-88 (Hebrew); R. de Vaux, *Fouilles de Khirbet Qumrân et de Ain Feshkha* (ed. J.B. Humbert and A. Chambon; Göttingen: Vandenhoeck & Ruprecht, 1994), pp. 192, 195-99, 342.

A dam was built under the second waterfall in the wadi, creating a small pool from which the aqueduct started. The dam deflected part of the running water (when there was a flood) from the wadi to the aqueduct. Forty metres beyond the waterfall the aqueduct becomes a subterranean tunnel 13.5 m long. After the tunnel, the aqueduct was built on supporting walls for another 130 m. On the plateau the aqueduct continues for about another 200 m, descending to Khirbet Qumran.

The Aqueducts of Hyrcania

Khirbet el-Mird, Hyrcania, is located 9 km west of Khirbet Qumran.⁴ Two water aqueducts brought water to Hyrcania:⁵ the northern one, which is 2 km long, started in a wadi north of the site (Fig. 6.3). It started above a waterfall of about 8 m in height. A dam about 4–5 m wide was built in a diagonal direction in order to deflect part of the water to the aqueduct. The width of this aqueduct in its upper part is about 1 m—double the width of the other aqueducts in the Judean desert. Thus, this aqueduct was designed to receive a vast quantity of flood water in a relatively short time.

The southern aqueduct is 9 km long, starting in Wadi Qidron (Wadi en-Nar). After 1300 m, when the wadi turns south, the aqueduct crosses the wadi and continues on the east side; in the area where in the Byzantine period the St Saba Monastery was built (see Fig. 6.3). The aqueduct leaves Wadi Qidron and changes its direction to the north-east, toward the fortress. The aqueduct then goes through three high bridges and a few small ones. The southern bridge is 7.5 m at its highest point. The highest point of the middle bridge was 17 m, while the northern bridge was 9 m above the wadi.

The two aqueducts meet about 750 m west of Hyrcania. Due to the brittle bedrock in this area, an open canal 500 m long was built. Here two impressive bridges were erected. West of the fortress stood another monumental bridge, through which the two aqueducts passed, bringing water to the cisterns (see Fig. 6.4).

4. J. Patrich, 'Hyrcania', in E. Stern (ed.), *The New Encyclopedia of Archeological Excavations in the Holy Land* (4 vols.; Jerusalem: Israel Exploration Society, 1993), II, pp. 639-41; C.R. Conder and H.H. Kitchener, *The Survey of Western Palestine*, III (London, 1883), p. 212; G.R.H. Wright, 'The Archaeological Remains at el-Mird in the Wilderness of Judaea', *Bib* 42 (1961), pp. 5-6 (5).

5. J. Patrich, 'The Aqueducts of Hyrcania', in Amit, Hirschfeld and Patrich (eds.), *Aqueducts of Ancient Palestine*, pp. 243-60.

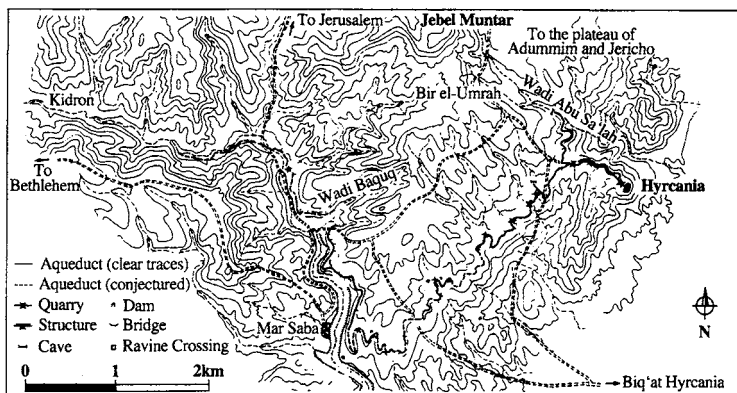


Fig. 6.3. *Map of the aqueducts of Hyrcania*
(Taken from Patrich, 'The Aqueducts of Hyrcania' [see n. 5])

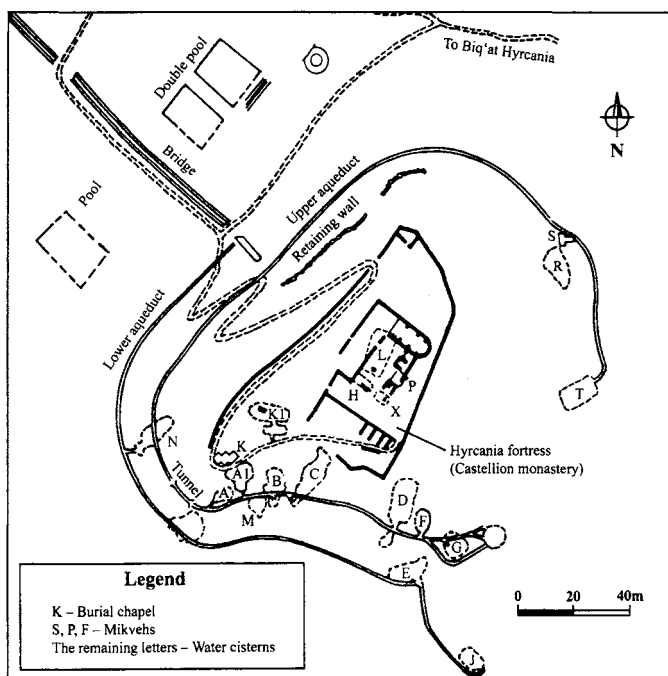


Fig. 6.4. *Map of Hyrcania and its vicinity*
(Taken from Patrich, 'Hyrcania' [see n. 4])

Two water pools were built north of the bridge (see Fig. 6.5). The northern pool measures $19 \times 18 \times 5$ m. The southern pool was not preserved as well as the northern one, but it was slightly smaller, its depth being 2.6 m. A third pool was built south of the bridge.

The aqueducts that carried water to Hyrcania filled 14 cisterns, 12 of which were dug in two lines south of the fortress—8 on the upper line and 4 on the lower level. Two additional cisterns were dug north-east of the fortress. The northern aqueduct was presumably built during the Hasmonaean period, filling the 4 cisterns that are located south of the fortress, on the lower level, while the southern aqueduct, which was built during the Herodian period, filled the other 10 cisterns—8 south of the fortress and 2 in the north-east. Three ritual baths were built near the cisterns (see Fig. 6.4).

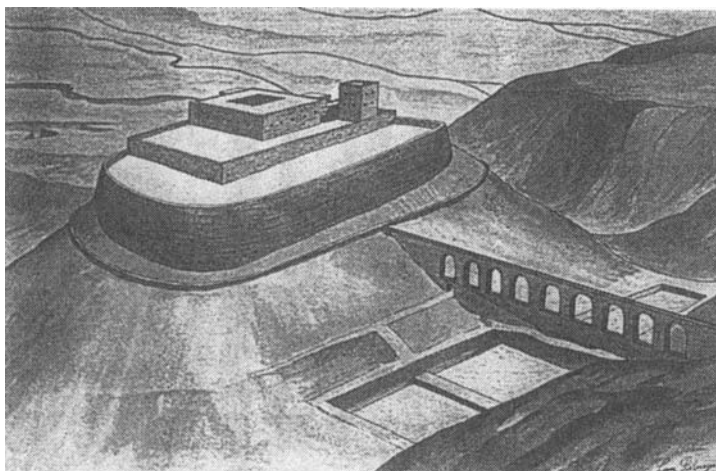


Fig. 6.5. *Northern bridge with the pools*
(Taken from Patrich, 'The Aqueducts of Hyrcania' [see n. 5])

The Aqueducts to Tell el-Aqabeh and to Jericho from Wadi el-Qelt

The fortress of Tell el-Aqabeh is located south of Wadi el-Qelt, 12 km north of Qumran.⁶ This fortress is usually identified with Cypros.⁷ Two

6. E. Netzer, 'Cypros', in Stern (ed.), *New Encyclopedia of Archaeological Excavations in the Holy Land*, I, pp. 315-17.

7. In 1981 we suggested identifying the fortress of Tell el-Aqabeh with Herodium of the hills on the Arabian frontier (Josephus, *War* 1.419); see H. Eshel and Y. Bin-

aqueducts carried water to the fortress of Tell el-Aqabeh (Fig. 6.6). The earlier one, dated to the Hasmonaeen period, was built around a hill west of the fortress. This aqueduct was c. 1 km long. It collected rainwater from this hill. During the Herodian period another aqueduct was built, carrying water from Wadi el-Qelt to Tell el-Aqabeh. This aqueduct, 14 km long, passed over a monumental bridge to the fortress (Fig. 6.7). Half a dozen additional bridges and three tunnels were built on the south side of Wadi el-Qelt. This aqueduct brought water to Tell el-Aqabeh, and from the fortress to the fields of the Royal Estate in Jericho.⁸

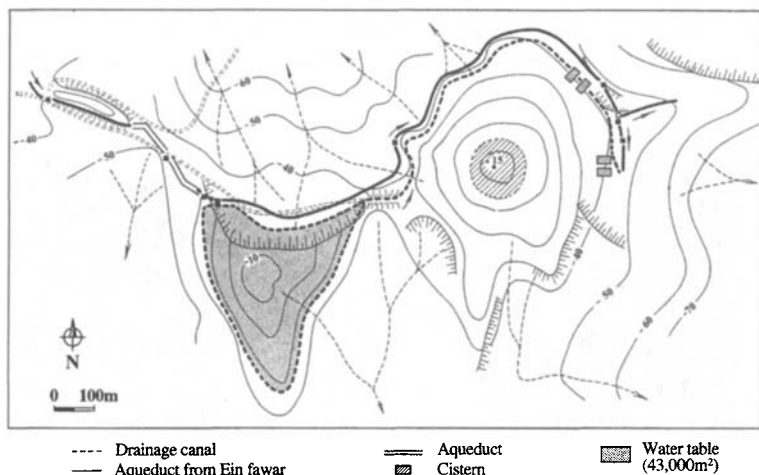


Fig. 6.6. Map of the aqueduct of Cypros (Tell el-Aqabeh)
(Taken from Garbrecht and Peleg, 'Die Wasserversorgung' [see n. 1])

Four cisterns were dug in the fortress of Tell el-Aqabeh, two north-east of the site and two east of the fortress. The northern cistern was probably

Nun, 'The Other Herodium and the Tomb of Herod', *Teva va-Aretz* 24 (1981), pp. 65-71 (Hebrew). This suggestion must be reconsidered because two fortresses from the Herodian period were found east of the Jordan. One of them may be identified as the Arabian Herodium; see A. Mallon, 'Deux fortresses au pied des Monts de Moab', *Bib* 14 (1933), pp. 400-407; K. Prag and H. Barnes, 'Three Fortresses on the Wadi Kafraïn, Jordan', *Levant* 28 (1996), pp. 41-61.

8. Conder and Kitchener, *Survey of Western Palestine*, pp. 190, 222, 227-28; Z. Meshel and D. Amit, 'Water Supply to Cypros Fortress', in Amit, Hirschfeld and Patrich (eds.), *Aqueducts of Ancient Palestine*, pp. 229-42 (Hebrew).

the smallest one.⁹ In the Herodian period the aqueduct carried water to the fortress of Tell el-Aqabeh all year long, and not only during certain days in the winter. Nevertheless, the cisterns were needed because an enemy could block the water in the aqueduct and deflect it at the beginning of a siege. Therefore, the soldiers guarding the fortress in the Herodian period had to ensure that the cisterns were always full.

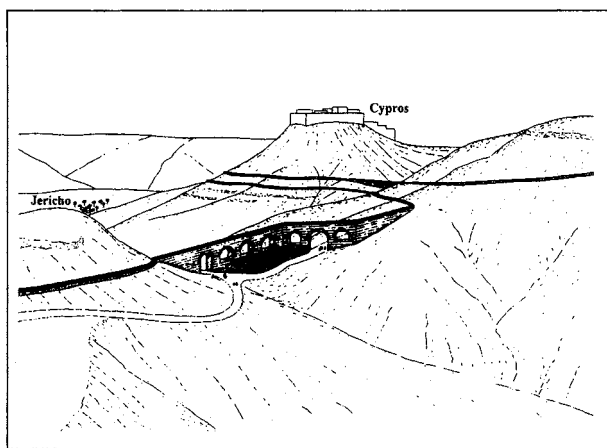


Fig. 6.7. *Drawing of the large bridge carrying the aqueduct leading to Cypros (Tell el-Aqabeh)*

(Taken from Meshel and Amit, 'Water Supply to Cypros Fortress' [see n. 8])

The earliest aqueduct that was built in the Judaeen Desert was built north of Wadi el-Qelt. This aqueduct carried water to the Hasmonaeen Palace and to the Royal Estate, which were built south of Jericho during the reign of John Hyrcanus I (134–104 BCE).¹⁰ It was 15 km long. King Herod built a 5 km aqueduct along the south side of the eastern part of Wadi el-Qelt, carrying some of the winter water of Wadi el-Qelt to the fields of Jericho. At the end of the Second Temple period, three aqueducts

9. This cistern has collapsed, but in the map of Meshel and Amit, 'Water Supply to Cypros Fortress', p. 234, the northern cistern is smaller than the other collapsed cistern beside it.

10. E. Netzer, 'The Hasmonaeen and Herodian Winter Palaces at Jericho', *IEJ* 25 (1975), pp. 89–100; *idem*, 'The Winter Palaces of the Judean Kings at Jericho at the End of the Second Temple Period', *BASOR* 228 (1977), pp. 1–14; *idem*, 'Tulul Abu el 'Alayiq', in Stern (ed.), *New Encyclopedia of Archaeological Excavations in the Holy Land*, IV, pp. 682–91; H. Eshel, 'The Historical Background of the Pesher Interpreting Joshua's Curse on the Rebuilder of Jericho', *RevQ* 15 (1992), pp. 409–20.

(one on the north side, one on the south side passing Tell el-Aqabeh and the third being a short aqueduct, on the south side of the eastern part) led the water of Wadi el-Qelt to the Royal vineyards at Jericho. During that period more than 34 km of aqueducts were built along Wadi el-Qelt.¹¹

The Aqueduct of Doq at Ras Qarantal

The fortress of Ras Qarantal is located west of Jericho, 16 km north of Qumran (Fig. 6.8). This fortress is identified with Doq or Dagon of the Hasmonaean Period.¹²

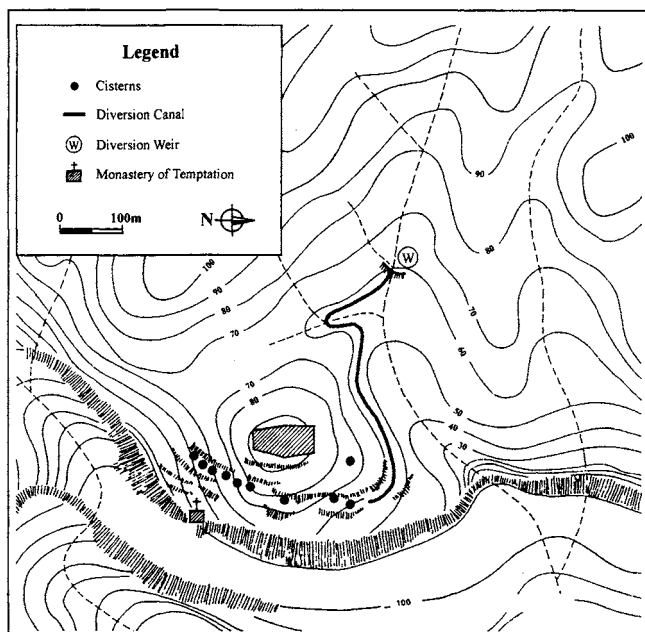


Fig. 6.8. *Map of the aqueduct of Doq*

(Taken from Garbrecht and Peleg, 'Water Supply of the Desert Fortresses' [n. 1])

11. E. Netzer, 'Water Channels and a Royal Estate from the late Hellenistic Period in the Western Plains of Jericho', *Leichtwiess-Institut für Wasserbau der Technischen Universität Braunschweig Mitteilungen Heft 82* (1984), pp. 1-12; G. Garbrecht and E. Netzer, *Die Wasserversorgung des geschichtlichen Jericho und seiner Könighchen Anlagen* (Braunschweig: Braunschweig Leichtwiess-Institut für Wasserbau der Technischen Universität Braunschweig, Mitteilungen 115, 1991).

12. See Tzafrir, 'Desert Fortresses', p. 122. We have suggested identifying this fortress as Cypros of the Herodian period as well; see Eshel and Bin-Nun, 'The Other Herodium'.

The aqueduct that was built in order to carry water to the fortress of Ras Qarantal measures c. 700 m in length.¹³ This aqueduct started at the foot of a small waterfall, running north of the fortress. It carried water to nine cisterns, carved east of the fortress.¹⁴ Most of the cisterns are rectangular and measure between 7.5 m and 14 m long and are 3.5 m to 5.5 m wide. The northern cistern, no. 1, is not bigger than the other cisterns. The content of all these cisterns is 2090 m³.

*Paragraphs in the Copper Scroll that
are Related to Water Aqueducts*

The Aqueduct of Sekakah

In the beginning of col. 5 of the Copper Scroll we read:

1 ברוש אמת המים [הבאה ל]¹⁵
 2 סככא מן הצפון תחת האבן¹⁶
 3 הגדולה חפור אמות [של]
 4 כסף כב 7

1. At the head of the water aqueduct [which penetrates to]
2. Sekakah¹⁷ from the north, bene[ath the] large
3. [stone,] dig for [three] cubits:
4. seven talents of silver.

As some scholars have already suggested, this aqueduct should be identified with the aqueduct of Wadi Qumran.¹⁸ The dam that was built in this wadi is mentioned before the aqueduct at the end of col. 4:

13. D. Amit, 'The Water System of Dok Fortress', in Amit, Hirschfeld and Patrich (eds.), *Aqueducts of Ancient Palestine*, pp. 223-28 (Hebrew).

14. O. Meinardus, 'Notes on the Laurae and Monasteries of the Wilderness of Judaea (III)', *LA* 19 (1969), pp. 325-26.

15. E. Puech orally suggested the reconstruction סככא [ממערבא ל] ברוש אמת המים, 'At the head of the water aqueduct coming from the west to Sekakah', which is possible.

16. G.B. Sarfati suggested (in a seminar) the reconstruction תחת הסככא. This reconstruction is based on 11.6. The term סככא is mentioned in the rabbinic literature, meaning 'a stone'; see *m. Kel.* 1.3; *b. Nid.* 69b; and E. Ben-Yehuda, *A Complete Dictionary of Ancient and Modern Hebrew* (Jerusalem: Thomas Yoseloff, 1959), p. 3124 n. 1.

17. The name of Qumran was Sekakah in the First and Second Temple periods; see H. Eshel, 'A Note on Joshua 15.61-62 and the Identification of the City of Salt', *IEJ* 45 (1995), pp. 37-40.

18. See J.T. Milik, 'Le rouleau de cuivre provenant de la grotte 3Q (3Q15)', in M.

13 ביגר של גי הסכבא חפור

14 אמת כסף כב 12

13. In the dam of Sekakah Gorge, dig

14. cubits: twelve talents of silver.

The word 'ג'ר' means 'a mound', as mentioned in Gen. 31.47, 'ג'ר שהדוהא'.¹⁹ It seems that 'ג'ר' in the Copper Scroll means 'a dam', where an aqueduct started.²⁰

It seems that the first treasure was hidden under a big stone on the plateau, north of the aqueduct, while the second treasure was hidden in the dam of Wadi Qumran.

The Aqueducts of Hyrcania

The dam where the southern aqueduct leading to Hyrcania began is mentioned in 8.8-9 of the Copper Scroll:

8 ביגר של פי צוק הקדרון

9 חפור אמות שלוש כב 7

8. In the dam of the Kidron cliff

9. dig for three cubits: seven talents of silver.

Therefore it seems that this treasure of seven talents of silver was hidden in the dam of Wadi Qidron (where the southern aqueduct of Hyrcania started).

In 4.3-5, an aqueduct is mentioned. Its destination was not preserved in the Copper Scroll. According to the context I suggest the reconstruction 'to Hyrcania' in line 3. This reconstruction is based on the assumption that there is some geographical order in the Copper Scroll. The next treasure was hidden in 'עמק עבון', which is probably 'עבון', mentioned in Josh. 7.24-26, to be identified with el-Buqa'a.²¹ The north aqueduct carrying

Baillet, J.T. Milik and R. de Vaux, *Les 'petites grottes' de Qumrân* (DJDJ III; Oxford: Clarendon Press, 1962), p. 263 n. 9; Ilan and Amit, 'Aqueduct of Qumran', p. 287. They also suggested that the 'ג'ר' is the dam in Wadi Qumran.

19. On the 'ג'ר סהדוהא' see J. Skinner, *Genesis* (ICC; Edinburgh: T. & T. Clark, 1910), p. 401. On 'ג'ר' in the Copper Scroll as a dam see B.-Z. Lurie, *The Copper Scroll from the Desert of Judah* (Jerusalem: Kiryat Sepher, 1963), p. 83 (Hebrew).

20. The word 'ג'ר' appears in 8.8, which will be discussed below, and in 6.14: 'שבמגזת הכוהן ביגר', although I do not understand this description, this 'ג'ר' can also be a dam of an aqueduct. See Lurie, *Copper Scroll*, p. 94.

21. See J.M. Allegro, *The Treasure of the Copper Scroll* (London: Routledge & Kegan Paul, 1960), pp. 64-68; Eshel, 'A Note on Joshua', pp. 37-38 n. 5. For a

water to Hyrcania passes the hills west of the Buqe'a. If we accept this restoration, we read:

3 באמא הבא[ה להורקניה] בביאתך
4 אמות ארבע[ין ואח]ת כסף
5 כב 55

3. In the aqueduct which go[es to Hyrcania] when you enter (the site)

4. fort[y-on]e cubits:

5. fifty-five talents of silver.

It seems, therefore, that we can identify this aqueduct as the north aqueduct, carrying water from the area west of the Buqe'a to Hyrcania. The treasure was hidden 41 cubits from the point where the trail met the north aqueduct west of Hyrcania.

The Aqueduct to Cypros or Another Reference to the Aqueduct to Hyrcania

In 7.3 the description of a treasure starts with the words באמא של קין, in the aqueduct of QY[. There are two possibilities to reconstruct this word. If we reconstruct קין[פרוס]—Cy[pros then we should read:

3 באמא של קין[פרוס]
4 האשוח הצפון[ני הגדול]
5 בארבע רוח[ות לשם]ולו
6 משח אמות עשרין [ואר]בע
7 בכרין ארבע מאות

3. In the aqueduct of Cy[pros],

4. the nor[th bi]g reservoir []

5. on the four si[des to its le]ft (or: to its north)²²

6. measure twenty-[fo]ur cubits:

7. four hundred talents.

It seems that the reconstruction ו[ל הגדול] at the end of line 4 is certain. The word משח means 'to measure', namely 'to measure twenty-four cubits, left (or north) of the four sides of the big northern reservoir'. The fortress of Cypros was named after King Herod's mother. If the reconstruction

discussion of Iron Age water irrigation systems in the Buqe'a, see F.M. Cross and J.T. Milik, 'Explorations in the Judaean Buqe'ah', *BASOR* 142 (1956), pp. 5-17; and L.E. Stager, 'Farming in the Judean Desert During the Iron Age', *BASOR* 221 (1976), pp. 145-58.

22. In biblical Hebrew and in Arabic שמאל means north; see BDB, p. 969.

באמא של קיןפרוס is accepted, then the Copper Scroll was composed later than the Herodian Period.

Most scholars identify Cypros with the fortress of Tell el-Aqabeh.²³ There are four cisterns in Tell el-Aqabeh. The north cistern collapsed, but there is no reason to assume that this cistern was bigger than the other three.²⁴

I suggested in 1981 that Cypros was the Herodian name of the fortress built at Ras Qarantal, which was named Doq or Dagon during the Hasmonaeen period.²⁵ If this is correct, then a problem arises, since in another description in the Copper Scroll, in col. 7, we read:

11 בדוק תחת פנת המשמרה
12 המנורחית הפור אמות שבע
22 ככ 13

11. In Doq,²⁶ under the eastern

12. guardhouse dig for seven cubits:

13. twenty-two talents of silver.

This description raises the question if it is possible that the same fortress be given different names in the Copper Scroll, one representing the Hasmonaeen name and the other the Herodian name.²⁷ Nevertheless the northern cistern at Ras Qarantal is not bigger than the other cisterns.²⁸ Therefore, we shall consider the possibility of reconstructing in 7.3 [...קיןדרון] באמא של קיןפרוס and reading, 'In the aqueduct of Qi[dron]'.²⁹

23. See Tzafrir, 'Desert Fortresses', p. 123.

24. See the map in Meshel and Amit, 'Water Supply to Cypros Fortress', p. 234 and Fig. 6.6 above, in which the north-eastern cistern is smaller than the other collapsed cistern beside it.

25. Eshel and Bin-Nun, 'The Other Herodium'.

26. For the identification of this treasure in the fortress of Ras Qarantal see Milik, DJDJ III, p. 265; Lurie, *Copper Scroll*, p. 97.

27. If we accept the prevailing suggestion to identify Cypros in Tell el-Aqabeh, we will face a similar problem, in light of Allegro's reasonable suggestion of identifying Mount Gerizim of 12.4 with Tell el-Aqabeh. See Allegro, *Treasure of the Copper Scroll*, pp. 75-76. On the polemic use of the toponym Mount Gerizim in the Copper Scroll, see H. Eshel, 'The Samaritans in the Persian and Hellenistic Periods: The Origins of Samaritanism' (PhD dissertation, Hebrew University Jerusalem, 1994), pp. 193-95 (Hebrew).

28. See the plans of these cisterns in Amit, 'Water System of Dok Fortress', p. 226 and Fig. 6.8 above.

29. I would like to thank Dr I. Knohl for this suggestion.

- 3 באמא של קינדרון...³⁰
 4 האשוח הצפוני הגדול
 5 בארבע רוחנות לשמןולו
 6 משח אמות עשרין ווארבע
 7 בכרין ארבע מאות

3. In the aqueduct of Qi[dron...,
 4. the nor[th bi]g reservoir []
 5. on the four si[des to its le]ft
 6. measure twenty-[fo]ur cubits:
 7. four hundred talents.

If we accept this reading, then this passage deals with the south aqueduct of Hyrcania and not with the aqueduct of Cyprus. One problem with this reconstruction is that קינדרון is written in col. 8 without a *yod*.³¹ Nevertheless this possibility fits the archaeological remains of Hyrcania, and if we accept it, we can identify the אשוח, 'the reservoir', with the big pool north of the bridge west of Hyrcania.³² Therefore, we may assume that this treasure was hidden 24 cubits north of the northern pool west of Hyrcania.

An Aqueduct at Wadi el-Qelt?

A hint of one of the aqueducts that was built in Wadi el-Qelt may be found in 7.14-16:

- 14 על פי יציאת המים של הכזיב
 15 בא חפור אמות שלוש עד הסור
 16 כב 60 זהב בכרין שתיים

14. By the mouth of the water outlet of the Koziba
 15. dig three cubits to the rock
 16. sixty talents of silver, two talents of gold.

30. We may consider reconstructing line 3:

- 3 באמא של קינדרון בבואה אל
 4 האשוח הצפוני הגדול

3. In the aqueduct of Qi[dron when it enters
 4. the nor[th bi]g reservoir

31. In some cases we can show that there is no consistency in the Copper Scroll. For example, sometimes the scribe wrote אמות המים, 'water aqueduct' (5.1), and sometimes he wrote only אבא, 'aqueduct' (4.3).

32. The word אשוח can be better understood to be a water pool than a cistern. See the bibliography in J. Hoftijzer and K. Jongeling, *Dictionary of the North-West Semitic Inscriptions* (Leiden: E.J. Brill, 1995), pp. 122-23. See Figs. 6.4 and 6.5 above.

Koziba is probably the name of the eastern part of Wadi el-Qelt since the name of the monastery built there in the Byzantine period was Choziba.³³ If this description describes an aqueduct, it should probably be identified as the beginning of one of the aqueducts built in Wadi el-Qelt.

Appendix: Two Unknown Aqueducts in the Copper Scroll

Two other water aqueducts are mentioned in the Copper Scroll, but I cannot suggest any identification for them. In 1.9-12 we read:

9 בתל של כחלת כלי בלגין ואפרון
10 הכל של דמע והאצר השבעי מעסר
11 שני כפי גל פתחו בשולי האמא מן הצפון
12 אמות שש עד נקרת המבילה XAF

9. In the ruins of Kohlit, tithe vessels in a flask container and gray
10. (silver coins?)³⁴ All tithe and stored Seventh-Year produce (and) Second
11. Tithe, in the mouth of the mound its opening is at the edge of the aqueduct on the north
12. six cubits to the crevice (used) for ritual baths XAF

Since little is known about Kohlit other than it was located in the desert, I cannot identify this aqueduct.³⁵ In the beginning of col. 8 we read:

1 בא[מא שבדרך המזרח בית
2 האוצר שבמזרח אחיה (?)
3 כלי דמע וספרין אל תדקם (?)

1. In the aqueduct which is on the road to the east of the store
2. house, to the east of Ahiya (?):
3. tithe-vessels and books do not appropriate them (?)

33. On Koziba, see Milik, DJDJ III, p. 265; B. Pixner, 'Unravelling the Copper Scroll Code: A Study on the Topography of 3Q15', *RevQ* 11(1983), pp. 323-65 (349 n. 27).

34. Coins that are called 'black silver' are mentioned in the Greek documents of Babatha. Therefore we may assume that אַפּוֹרִן, 'Grays', are some kind of silver coins. See Y. Meshorer, 'The "Black Silver" Coins of the Babatha Papyri: A Re-evaluation', *Israel Museum Journal* 10 (1992), pp. 67-74; N. Lewis, 'Again, the Money Called Black', in R. Katsoff (ed.), *Classical Studies in Honor of David Sohlberg* (Ramat Gan: Bar-Ilan University Press, 1996), pp. 399-401.

35. On Kohlit in the desert, see *b. Qid.* 66a. I see no reason to accept B. Pixner's suggestion that Kohlit is the monastic centre of an Essene settlement (Pixner, 'Unravelling the Copper Scroll Code', p. 337). See now B. Zissu, 'The Identification of the Copper Scroll's *Kahelet* at 'Ein Samiya in the Samaritan Desert', *PEQ* 133 (2001), pp. 145-58; and there is an aqueduct at 'Ein Samiya.

No fortress or site by the name Beth-Achzor is known in the Judaean Desert, and although the aqueduct to Tell el-Aqabeh passes near the Roman road, I cannot identify this aqueduct.

Summary

Several water aqueducts are mentioned in the Copper Scroll. The water aqueduct that led water to Sekakah is to be identified with the aqueduct of Wadi Qumran. Of the other aqueducts in the area of Qumran two projects were long and notable; one carrying water to Hyrcania, and the other built along Wadi el-Qelt. It seems that those two projects are mentioned in the Copper Scroll.

There is evidence that the author of the Copper Scroll was familiar with the Judaean Desert and used the aqueducts of the Second Temple period as reference points for the hidden treasure.³⁶ It is important to note that although the fortresses were occupied by soldiers at all times, the aqueducts leading to those fortresses were in open areas, and everybody had access to them.

36. Without discussing the disagreements about the authenticity of the Copper Scroll, I would like to note that the only treasure found in Israel that might be connected to the temple, was found in 1960 at Isfiya (or Ussfiya) on Mount Carmel. This hoard is of about 4500 silver coins (3400 Tyrian shekels, about 1000 half-shekels and 160 Roman denarii of Augustus). It was suggested that this hoard is undelivered temple contributions due to the military conflict between the Romans and the Jews in Galilee in 67 CE. See L. Kadman, 'Temple Dues and Currency in Ancient Palestine in the Light of Recent Discovered Coins-Hoards', *Israel Numismatic Bulletin* 1 (1962), pp. 9-11; L. Kadman, 'Temple Dues and Currency in Ancient Palestine in the Light of Recent Discovered Coins-Hoards', in *Congresso Internazionale di Numismatica, 11-16 September 1961, Rome*, II (Roma: Istituto italiano di Numismatica, 1965), pp. 69-76; D. Sperber, 'Numismatics and Halacha', *Israel Numismatic Journal* 2 (1964), pp. 16-18.